Michael Lulic

Results

Q1)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    int numOfItems;
    double priceItem, taxRate;
    double subtotal, taxAmount, total;

    cout << "Enter the number of items: ";
    cin >> numOfItems;

    cout << "Enter the price per item: ";
    cin >> priceItem;

    cout << "Enter the tax rate (%): ";
    cin >> taxRate;

    subtotal = numOfItems * priceItem;
    taxAmount = subtotal * (taxRate/100);
    total = subtotal + taxAmount;

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    cout << "Subtotal: $" << subtotal << endl;
    cout << "Tax: $" << taxAmount << endl;
    cout << "Total: $" << total << endl;

    return 0;
}
```

```
Enter the number of items: 3
Enter the price per item: 20
Enter the tax rate (%): 15
Subtotal: $60.00
Tax: $9.00
Total: $69.00
```

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Q2)

```cpp
#include <iostream>

using namespace std;

int main(){
    int secondsEntered, sec, min, hour;
    double minAsdouble;

    cout << "Enter the duration in seconds: ";
    cin >> secondsEntered;

    sec = secondsEntered;

    hour = sec / (60*60);
    sec = sec % (60*60);

    min = sec / 60;
    sec = sec % 60;

    minAsdouble = (double) secondsEntered / 60;

    cout << "Hours: " << hour << endl;
    cout << "Minutes: " << min << endl;
    cout << "Seconds: " << sec << endl;
    cout << "Total minutes (double): " << minAsdouble;

    return 0;
}
```

```
Enter the duration in seconds: 3670
Hours: 1
Minutes: 1
Seconds: 10
Total minutes (double): 61.1667
```

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Q3)

```cpp
#include <iostream>
#include <cmath>

int main(){
    float x = 2;
    float y = 2;
    float z = 3;


    x = x + (((-1 * y) + (4 * x)) / (2 * z)) - (((x * sqrt(y)) + 6) / 4);

    std::cout << x;
    return 0;
}
```

```
0.792893
```

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Q4)

```cpp
#include <iostream>

using namespace std;

int main(){
    int userNum1, userNum2;
    int difference, divisionResult, remainder;

    cout << "Enter the first number: ";
    cin >> userNum1;

    cout << "Enter the second number: ";
    cin >> userNum2;

    //Calculates if num1 can divide num2
    if (userNum2 != 0){
        if (userNum1/userNum2 != 0 && userNum1%userNum2 == 0) {
            cout << userNum1 << " is divisible by " << userNum2 << "." <<endl;
        }
        else {
            cout << userNum1 << " is not divisible by " << userNum2 << "." <<endl;
        }
    }
    else {
        cout << "Cannot divide by 0. " << userNum1 << " is not divisible by " << userNum2 << "." <<endl;
    }

    //Calculates if num1 can divide num2
    if(userNum1 != 0){
        if (userNum2/userNum1 != 0 && userNum2%userNum1 == 0) {
            cout << userNum2 << " is divisible by " << userNum1 << "." <<endl;
        }
        else {
            cout << userNum2 << " is not divisible by " << userNum1 << "." <<endl;
        }
    }
    else {
        cout << "Cannot divide by 0. " << userNum2 << " is not divisible by " << userNum1 << "." <<endl;
    }

    //Absolute difference
    difference = userNum1 - userNum2;

    if (difference < 0){
        difference *= -1;
    }

    cout << "The absolute difference between the two numbers is " << difference << endl;
```

```cpp

    //Divide num1 by num2 and show remainder
    if (userNum2 != 0){
        divisionResult = userNum1 / userNum2;
        remainder =  userNum1 % userNum2;

        cout << "The result of dividing " << userNum1 << " by " << userNum2 << " is " << divisionResult << " with a remainder of " << remainder <<endl;
    }
    else {
        cout << "Cannot divide by 0. " << userNum1 << " is not divisible by " << userNum2 << " and has no remainder." <<endl;
    }

    return 0;
}
```

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```
Enter the first number: 15
Enter the second number: 6
15 is not divisible by 6.
6 is not divisible by 15.
The absolute difference between the two numbers is 9
The result of dividing 15 by 6 is 2 with a remainder of 3
```

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Q5)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    double userIncome;
    double tax, finalIncome;

    cout << "Enter your annual salary: ";
    cin >> userIncome;

    // 10k
    if (userIncome < 10000){
        tax = 0;

        finalIncome = userIncome;
    }

    // 10 - 20k
    else if (userIncome >= 10000 && userIncome <= 20000){
        tax = (userIncome - 10000) * 0.1;
        tax -= 500;

        if (tax < 0){
            tax = 0;
        }

        finalIncome = userIncome - tax;
    }


    // 20 - 50k
    else if (userIncome > 20000 && userIncome <= 50000){
        tax = ((userIncome - 20000) * 0.2) + 1000;
        tax -= 1000;

        if (tax < 0){
            tax = 0;
        }

        finalIncome = userIncome - tax;
    }

```

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```
    // 50 - 100k
    else if (userIncome > 50000 && userIncome <= 100000){
        tax = ((userIncome - 50000) * 0.3) + 7000;
        tax -= 500;

        if (tax < 0){
            tax = 0;
        }

        finalIncome = userIncome - tax;
    }

    // 100k +
    else{
        tax = ((userIncome - 100000) * 0.4) + 22000;

        finalIncome = userIncome - tax;
    }

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    cout << "Salary: $" << userIncome << endl;
    cout << "Tax: $" << tax << endl;
    cout << "Salaty after tax: $" << finalIncome << endl;


    return 0;
}
```

```
Enter your annual salary: 10000
Salary: $10000.00
Tax: $0.00
Salaty after tax: $10000.00
```

```
Enter your annual salary: 25300
Salary: $25300.00
Tax: $1060.00
Salaty after tax: $24240.00
```

```
Enter your annual salary: 155000
Salary: $155000.00
Tax: $44000.00
Salaty after tax: $111000.00
```

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Q6)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    double weight, length, width, height;
    double dimensionalWeight, finalWeight, cost;

    cout << "Enter the package weight in kilograms: ";
    cin >> weight;

    cout << "Enter the package dimensions (length, width, height) in centimeters: ";
    cin >> length;
    cin >> width;
    cin >> height;

    dimensionalWeight = (length * width * height);
    dimensionalWeight /= 5000;

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    cout << "The dimensional weight is " << dimensionalWeight << " kg." << endl;

    if (dimensionalWeight > weight) {
        finalWeight = dimensionalWeight;
    }
    else {
        finalWeight = weight;
    }

    if (finalWeight <= 2){
        cost = 5;
    }
    else if (finalWeight > 2 && finalWeight <= 10) {
        cost = 10;
    }
    else if (finalWeight > 10 && finalWeight <= 20) {
        cost = 20;
    }
    else {
        cost = -1;
    }

    if (cost > 0){
        cout << "The shipping cost is $" << cost;
    }
```

```cpp
    else {
        cout << "The package is too heavy to be shipped.";
    }

    return 0;
}
```

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```
Enter the package weight in kilograms: 3
Enter the package dimensions (length, width, height) in centimeters: 50 40 30
The dimensional weight is 12.00 kg.
The shipping cost is $20.00
```

```
Enter the package weight in kilograms: 1.5
Enter the package dimensions (length, width, height) in centimeters: 20 10 5
The dimensional weight is 0.20 kg.
The shipping cost is $5.00
```

```
Enter the package weight in kilograms: 25
Enter the package dimensions (length, width, height) in centimeters: 60 50 40
The dimensional weight is 24.00 kg.
The package is too heavy to be shipped.
```

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Q7)

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    int optionEntered;
    double userValue;
    double convertedValue;

    cout << "Enter the value you want to convert: ";
    cin >> userValue;

    cout << "Select the conversion:" << endl;
    cout << "1. Kilometers to Miles" << endl;
    cout << "2. Pounds to Kilograms" << endl;
    cout << "3. Celcius to Fahrenheit" << endl;
    cout << "4. Liters to Gallons" << endl;
    cout << "Enter your choice (1-4): ";
    cin >> optionEntered;

    //Forces 2 decimal places
    cout << fixed << setprecision(2);

    switch (optionEntered){
        case 1:
            convertedValue = userValue * 0.621371;
            cout << userValue << " kilometers " << "is equal to " << convertedValue << " miles.";
            break;
        case 2:
            convertedValue = userValue * 0.453592;
            cout << userValue << " pounds " << "is equal to " << convertedValue << " kilograms.";
            break;
        case 3:
            convertedValue = (userValue * 9 / 5) + 32;
            cout << userValue << " celcius " << "is equal to " << convertedValue << " fahrenheit.";
            break;
        case 4:
            convertedValue = userValue * 0.264172;
            cout << userValue << " liters " << "is equal to " << convertedValue << " gallons.";
            break;
        default:
            cout << "Invalid selection.";
            break;
    }

    return 0;
}
```

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```
Enter the value you want to convert: 100
Select the conversion:
1. Kilometers to Miles
2. Pounds to Kilograms
3. Celcius to Fahrenheit
4. Liters to Gallons
Enter your choice (1-4): 1
100.00 kilometers is equal to 62.14 miles.
```

```
Enter the value you want to convert: 50
Select the conversion:
1. Kilometers to Miles
2. Pounds to Kilograms
3. Celcius to Fahrenheit
4. Liters to Gallons
Enter your choice (1-4): 5
Invalid selection.
```